

Polynomial conditioning of the cubic C1 spline Schrödinger Toeplitz family

Difficulty: hard **Importance:** interesting to specialist

Status: [Lean verified](#)

Topic: Block Toeplitz systems and discretization stability

Rating rationale: Historical ratings retained. The fixed explicit band structure makes this a hard asymptotic question, but available symbol criteria leave the relevant regime undecided; its immediate importance is to specialists in structured discretization stability.

Resolution: affirmative, 11 September 2026

George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, proves that for every fixed $\rho > 0$ there are $K_\rho > 0$ and n_ρ such that

$$\kappa_2(H_n(\rho)) \leq K_\rho n \quad (n \geq n_\rho).$$

Thus the original question is answered affirmatively with exponent $\alpha_\rho = 1$. The result includes eventual invertibility, the parameter $\rho = \sqrt{10}$, and the exact uncorrected Toeplitz boundary entries. The original ID, full statement, and historical ratings are retained below.

Formal scope: The Lean proof below certifies the original polynomial target with exponent two; it does not certify this stronger linear estimate.

The precise locator is the **Theorem in Section 1, proved in Sections 2–4** of the [complete manuscript](#). [Proof PDF](#) · [Standalone TeX](#). The proof derives a four-state recurrence, excludes all numerator cancellation, and bounds its finite Green matrix to obtain a linear inverse bound.

A separate [independent Codex-agent mathematical review](#) returned **PASS** for the complete target. The [submission record](#) documents substantial ChatGPT/Codex assistance, exact symbolic checks, the public branch/fork audit, and verification limits. That original review was informal automated-agent review. The separate Lean proof of the complete original target is documented below; external human peer review is not claimed. The source authors retain credit for the family, the root classification and the question.

Lean proof and verification evidence

Formalization: George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology. The [complete Lean proof at immutable revision c701bfee](#) proves the full original target with exponent **two**:

$$\forall \rho > 0 \exists K_\rho > 0, n_0 \geq 1 : \det H_n(\rho) \neq 0, \quad \kappa_2(H_n(\rho)) \leq K_\rho n^2 \quad (n \geq n_0).$$

The formal result uses the literal complex $2n \times 2n$ pure Toeplitz matrix, all eight original coefficient blocks, the exact boundary entries and the genuine Euclidean operator norm, with infinite condition number for singular matrices. It includes $\rho = \sqrt{10}$. The original question permits any nonnegative polynomial exponent, so this settles it completely. **The stronger linear bound in the retained informal manuscript is outside the formalized claim.**

All **22 declarations** are listed in [formalization.yaml](#). The [quadratic estimate and final theorem](#) conclude with *polynomial_conditioning* in namespace *NLA.MF22*. The independently reviewed [Challenge, definitions](#) and [source correspondence](#) preserve the complete target. Exact fixed-size transfer algebra, root classification, noncancellation and Green reconstruction prove an actual inverse. Cancelling growing terms before estimates and using dimension-times-entry norm bounds gives exponent two. A completed square and a consumed LeanCert integer-positivity certificate avoid interval searches.

The project pins Lean **4.33.1**, Mathlib [0df444a360eaa60ab8c11dca51a86af692955474](#) and LeanCert [621a43d7cf21f87872392a01e874f2f1dbddc926](#). On **16 September 2026 UTC**, the [actual GitHub Linux job](#) checked the exact proof revision above. All 22 targets passed LeanCert kernel-trust assertions, non-root sandboxed Comparator statement matching and Lean default-kernel replay. Their transitive axioms are only *propxet*, *Classical.choice* and *Quot.sound*. Required kernel, statement-mismatch, forbidden-axiom and sandbox controls passed inside that job. The [retained actual logs, ZIP and input hashes](#) bind its execution to all **634** candidate project files.

Two independent AI-agent referees accepted the complete mathematical source and actual canonical evidence: the [continuation referee](#), with a separate full 29-file review, and the [root referee](#). They applied the repository’s scoped Tau Ceti protocol. Their reviews are distinct from external human peer review or proof of checker-software infallibility. No local macOS Lean execution is claimed. Bogoya, Böttcher, Ferrari, Grudsky and Serra-Capizzano retain credit for the family, source root classification and original question.

For an ordinary proof build, enter [the Lean project](#) with its pinned toolchain and run:

```
lake exe cache get && lake build
```

The default target builds the complete Solution without importing Challenge’s specification placeholders. For the additional Comparator and rejection checks, follow the [shared Linux instructions](#). The evidence above names the checked proof commit; later publication and merge commits require separate exact-commit checks.

The [bounded public duplicate audit](#) checked 249 public branch heads across 14 repositories and 149 upstream PR records. Its only MF-22 Lean hits were this campaign’s development and standalone proof commits. This is not a claim about private, deleted or otherwise undiscoverable work, nor a new literature search.

Statement

Define real 2×2 matrices by

$$\begin{aligned} B_{-1} &= \frac{3}{40} \begin{pmatrix} -1 & 0 \\ -5 & 0 \end{pmatrix}, & B_0 &= \frac{3}{40} \begin{pmatrix} 0 & -5 \\ 16 & -5 \end{pmatrix}, \\ B_1 &= \frac{3}{40} \begin{pmatrix} -5 & 16 \\ -5 & 0 \end{pmatrix}, & B_2 &= \frac{3}{40} \begin{pmatrix} 0 & -5 \\ 0 & -1 \end{pmatrix}, \\ C_{-1} &= \frac{1}{80} \begin{pmatrix} 1 & 0 \\ 7 & 0 \end{pmatrix}, & C_0 &= \frac{1}{80} \begin{pmatrix} 24 & 7 \\ 0 & 25 \end{pmatrix}, \\ C_1 &= \frac{1}{80} \begin{pmatrix} -25 & 0 \\ -7 & -24 \end{pmatrix}, & C_2 &= \frac{1}{80} \begin{pmatrix} 0 & -7 \\ 0 & -1 \end{pmatrix}. \end{aligned}$$

Set $B_k = C_k = 0$ for all other integer indices. For a real parameter $\rho > 0$ and $n \geq 1$, form the $2n \times 2n$ complex block Toeplitz matrix

$$H_n(\rho) = (iB_{j-k} - \rho C_{j-k})_{j,k=0}^{n-1}, \quad i^2 = -1.$$

Use the spectral condition number $\kappa_2(H) = \|H\|_2 \|H^{-1}\|_2$ for nonsingular H and $\kappa_2(H) = +\infty$ for singular H .

Question (explicit asymptotic formulation of the source's polynomial-growth question). For every fixed $\rho > 0$, do there exist constants $K_\rho > 0$, $\alpha_\rho \geq 0$, and $n_\rho \in \mathbb{N}$, independent of n , such that

$$\kappa_2(H_n(\rho)) \leq K_\rho n^{\alpha_\rho} \quad \text{for every } n \geq n_\rho?$$

The parameter remains fixed as $n \rightarrow \infty$; no uniform-in- ρ bound or particular exponent is asserted. Eventual invertibility is part of the question. This is the exact pure Toeplitz family denoted $\tilde{\mathbf{S}}_n^{(3,1)}(\rho)$ in the source; n here counts its 2×2 blocks. It has no additional corner corrections.

Numerical significance

These matrices arise from cubic splines with C^1 continuity in time for a space-time Schrödinger discretization. The source proves a determinant-root classification of the associated polynomial symbol but expressly leaves polynomial growth of these condition numbers open. The linear bound recorded above explains the numerical stability seen for this intermediate spline regularity.

References and status check

- M. Bogoya, A. Böttcher, M. Ferrari, S. M. Grudsky and S. Serra-Capizzano, *Condition numbers of block Toeplitz matrices and stability of space-time IgA approximations for the wave and Schrödinger equations*, [arXiv:2608.24151v1](#), 25 August 2026. **Full text:** equation (5.22) gives B_k ; §5.3, equation (5.32) and the following coefficient display give the family and C_k ; the paragraph after Figure 9 and Remark 4.3 state the unresolved polynomial-growth issue.
- M. Ferrari and S. Gómez, *A matrix-based approach to the stability of a space-time isogeometric method for the linear Schrödinger equation*, [arXiv:2506.18859v2](#) (5 May 2026), abstract and introduction. Its analyzed method uses splines of maximal regularity, a different case from cubic C^1 splines.

On 2026-09-10, checked the complete August 2026 source, its version history, and targeted identifier/title/correction searches. Only v1 was listed, and no later proof or counterexample was found. This short, bounded follow-up search does not establish exhaustive openness. The source's more general claim involving only determinant-root counts is not needed for this particular, explicitly stated family.

The 2026-09-11 submission audit checked five public repositories and all 30 public branch heads, where MF-22 remained Open, and found no prior full solution in the checked documents or discussions. The [dated audit](#) records the scope and its limits. The source still listed only v1.